Q4. Illustrate Git architecture and discuss commonly used Git commands for repository creation, file management, branching, and merging. (16 Mark)

1. Introduction

Git is a distributed version control system used to track changes in source code and manage software projects. It allows developers to maintain different versions of a project, work on different features independently, and combine changes efficiently.

Git mainly consists of four important areas:



- **Working Directory**

The Working Directory is the folder where the developer creates and modifies project files.

- **Staging Area**

The Staging Area contains the changes selected for the next commit. The `git add` command moves changes from the Working Directory to the Staging Area.

- **Local Repository**

The Local Repository stores the committed versions of the project on the developer's computer. The `git commit` command saves changes into the repository.

- **Remote Repository**

The Remote Repository is stored on a remote server such as GitHub, GitLab, or Bitbucket. The `git push` command uploads local commits to the remote repository.

- **Repository Creation Commands**

1. Creates a new Git repository in the current directory. - **`git init`**

2. Creates a local copy of an existing remote repository- **`git clone`**

Example

```
git clone https://github.com/user/project.git
```

3. Connects the local repository to a remote repository - **`git remote add`**

Example

```
Git remote add origin
https://github.com/user/project.git
```

File Management Commands

`git init`	Creates a new repository
`git clone`	Copies an existing repository
`git status`	Shows file status
`git add`	Adds changes to the Staging Area
`git commit`	Saves changes to the Local Repository
`git log`	Displays commit history
`git branch`	Creates/displays branches
`git switch`	Changes the current branch
`git merge`	Combines two branches
`git rm`	Removes a file

5. Branching Commands

A **branch** is an independent line of development. Branching allows developers to work on a new feature without affecting the main branch.

- Displays the existing branches.

```
git branch
```

- **Create a new branch**

```
git branch feature
```

- **Switch to a branch**

```
git switch feature
```

- **Create and switch to a new branch**

```
git switch -c feature
```

Example:

```
git switch -c login-feature
```

The developer can now develop the login feature separately from the main branch.

6. Merging Commands

Merging is the process of combining the changes from one branch into another branch.

- First, switch to the main branch:

```
git switch main
```

- Then merge the feature branch:

```
git merge feature
```

The changes made in the `feature` branch are combined with the `main` branch.

Merge Conflict

A merge conflict occurs when two branches modify the same part of a file differently. Git identifies the conflict, and the developer must manually resolve it before completing the merge.

After resolving the conflict:

```
git add .
git commit -m "Resolved merge conflict"
```